AI100 Section 002 Midterm Project

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Introduction

This project uses ChatGPT's LLM capabilities with deep learning to determine if a country can be classified as high-income or not, through a binary classification problem. This project follows through with my Homework#1's prompt of calculating a high-income score based on different economic and social factors. I used the previous assignment's prompt while also giving the LLM additional requirements to use deep learning in producing a binary output.

Problem Definition

Predict whether a country is classified as high-income based on different economic and social determinants.

Dataset Curation

There were many social and economic metrics used to determine a country's income status: GDP in billions of US dollars, GDP per capita in US dollars, GDP growth rate, Population in millions, Inflation rate, Unemployment percentage, Life expectancy, Internet usage percentage.

These metrics were generated for each country by the LLM and then used to create a high-income classification label. The countries with higher weights of the metrics discussed above were then calculated and a binary value of either 1 (high-income) or 0 (not high-income) was assigned for each country.

Comma-Separated Value Dataset

country	gdp_usd_b	gdp_per_capit	gdp_growth_p	population_m	inflation_per	unemployment_p	life_expect	internet_users_p
Afghanistan	9738.67	90357.29	5.05	868.06	2.34	5.52	52.03	87.09
Albania	15629.39	67500.44	-2.77	1406.37	12.49	7.16	56.36	26.32
Algeria	7910.99	50232.06	1.75	422.29	9.18	5.05	60.23	42.61
Andorra	11858.36	74763.58	-0.8	745.64	8.89	2.35	71.26	25.18
Angola	1692.28	90185.02	7.62	1172.18	4.57	3.83	73.95	49.17
Antigua and Barbuda	3173.87	47445.66	-2.62	1318.52	3.88	20.21	60.91	56.29
Argentina	14214.92	18213.29	7.67	1123.94	14.09	26.95	70.93	92.05
Armenia	2301.72	19261.59	-2.5	471.74	5.83	8.87	79.01	41.75
Australia	7305.02	51921.97	-1.45	1163.19	1.12	29.62	77.03	27.69
Austria	144.57	77616.47	4.78	1057.06	11.57	3.15	62.55	20.31
Azerbaijan	22440.83	59514.68	0.64	92.17	4.66	10.43	75.54	66.74
Bahamas	23067.64	45282.65	-1.68	1034.21	11.41	17.28	76.98	53.95
Bahrain	13591.53	41074.36	-2.72	156.45	0.47	19.46	61	55.26
Bangladesh	23596.82	24283.33	1.51	1095.55	3.43	3.23	60.14	24.35
Barbados	24172.21	76924.94	3.97	1263.62	12.06	6.41	81.24	58
Belarus	20993.64	85211.8	0.5	159.58	3.42	13.39	78.63	86.61
Belgium	181.75	48912.4	1.59	322.06	1.8	10.79	83	38.77
Belize	13489.04	67024.39	1	1409.08	14.44	8.3	67.4	36.78
Benin	7406.57	4274.75	3.71	728.89	0.77	9.08	81.79	31.32
Bhutan	3768.12	46906.45	7.84	350.99	10.08	23.09	58.32	74.81
Bolivia	9562.99	60363.21	3.97	776.88	1.35	25.22	61.23	26.6
Bosnia and Herzegovina	1061.11	56462.12	4.45	24.06	7.68	7.57	72.58	25.52
Botswana	17964.69	37230.47	7.3	199.41	5.12	4.29	82.36	88.08
Brazil	6707.22	62970.5	5.99	805.05	7.94	8.01	53.26	89.85
Brunei	23410.97	60438.16	0.73	506.36	10.89	27.02	81.05	79.41
Bulgaria	16693.18	8725.98	-1.22	1302.9	9.1	1.27	53.55	69.05
Burkina Faso	132.6	15948.12	3.04	1003.25	9.78	7.5	74.93	31.12
Burundi	8461.07	71119.49	4.15	1231.38	9.86	17.48	53.28	42.73
Cabo Verde	6896	23783.82	7.7	570	13.38	19.3	77.82	54.73
Cambodia	14999.92	47195.17	-0.85	1047.56	4.21	1.71	72.59	25.76
Cameroon	24451.98	90660.07	7.06	536.74	0.23	27.92	64.99	96.03
Canada	25054.16	81153.49	0.24	558.4	12.77	10.19	55.93	59.56
Central African Republic	24340.09	66366.01	3.27	140.91	9.23	29.71	54.9	56.13
Chad	22811.82	70580.4	4.67	1018.6	5.39	9.51	78.33	82.1
Chile	22544.01	86827.26	2.62	727.2	11.97	19.85	74.57	80.83
China	23140.25	32639.14	1.13	136.28	8.67	2.04	66.3	58.3
Colombia	7450.79	56456.49	-2.66	54.16	12.34	11.45	54.45	56.48
Comoros	20020.06	21130.34	3.85	123.76	0.78	16.41	68.92	66.73
Congo	18878.65	92725.27	2.68	468.29	11.93	8.85	65.36	16.98
Costa Rica	660.09	91481.48	6.2	1009.17	6.13	6.03	55.48	32.27
Croatia	14280.34	68114.94	4.26	405.91	14.32	22.4	69.4	64.44
Cuba	10910.18	24136.26	0.92	1098.88	0.22	4.37	51.61	13.62
Cyprus	22242.12	67084.57	2.22	141.87	7.37	14.73	56.06	48.61
Czech Republic	10361.72	58813.08	3.99	65.7	5.62	19.15	67.61	86.23
Democratic Republic of the C	17126.38	16148.42	-2.22	931.51	0.4	17.99	82.91	61.22
Denmark	10093.03	61397.75	2.04	791.15	14.12	12.2	83.64	90.58
Djibouti	5091.37	7333.83	-1.89	26.43	1.42	20.81	52.49	38.39

Dominica	21966.91	2992.22	5.96	408.7	1.77	21.21	72.01	88.1
Dominican Republic	19112.11	76487.9	0.1	257.3	11.26	24.4	84.67	46.72
Ecuador	9673.1	73938.1	0.75	1349.6	12.88	13.44	76.28	77.15
Egypt	2682.12	85820.48	2.56	1198.37	4.8	26.97	63.62	10.96
El Salvador	23540.03	9399.2	0.51	1377.59	14.26	17.63	72.11	49.91
Equatorial Guinea	7624.19	31760.2	4.4	1090.95	11.87	23.9	53.19	54
Eritrea	1497.47	52565.62	1.86	1287.17	5.26	4.39	55	77.77
Estonia	16074.05	10325.76	-2.07	1016.41	1.09	24.83	74.72	17.24
Eswatini	2206.7	93741.45	1.12	537.44	12.19	28.47	84.51	77.05
Ethiopia	9783.37	8665.77	5.55	809.69	6.36	27.28	53.89	53.84
Fiji	296.18	44947.83	-2.38	172.29	1.76	19.83	76.11	61.92
Finland	25016.52	36112.81	0.14	1259.47	3.35	28.93	50.43	96.32
France	1123.11	84745.68	2.8	1439.8	1.11	17.06	83.93	56.56
Gabon	16364.74	66339.53	2	909.96	8.76	27.13	51.59	35.01
Gambia	24710.75	84662.85	2.01	899.2	4.16	6.46	66.23	41.45
Georgia	15175.48	8122.6	7.72	1430.01	10.47	16.55	60.83	82.43
Germany	17803.33	16118.52	7.02	1192.68	14.25	22.05	71.47	47.22
Ghana	24251.01	82383.22	-2.5	38.24	5.65	24.51	84.55	23.39
Greece	15447.8	36679.92	7.67	1221.07	12.57	14.59	64.52	34.33
Grenada	1466.71	82256.85	5.94	1449.59	14.95	17.11	76.91	94.08
Guatemala	22090.98	24100.19	1.96	187.29	14.31	18.58	58	69.78
Guinea	16071.72	34538.93	-1.75	973.78	7.8	23.4	68.21	85.84
Guinea-Bissau	14350.03	53640.36	6.64	585.06	2.01	1.83	76.43	65.21
Guyana	18306.37	20861.22	-1.5	21.1	5.26	18.11	63.73	48.94
Haiti	23508.22	33605.66	2.65	1136.3	5.95	19.04	80.18	94.51
Honduras	3824.76	88084.55	2.41	374.46	6.89	29.42	67.24	39.26
Hungary	16468.79	23421.72	-2.17	186.88	1.92	5.41	54.86	67.04
Iceland	4729.7	33361.86	6.86	687.25	10.01	6	56.73	13.64
India	4393.14	27043.21	-1.05	128.63	1.81	14.36	57.22	42.42
Indonesia	13089.35	65835.19	-2.57	1159.15	9.42	3.37	80.58	91.96
Iran	1588.97	26881.87	5.87	1084.98	2.77	7.07	62.97	53.12
Iraq	16075.01	35551.66	2.09	1083.84	0.55	8.32	74.97	89.67
Ireland	13304.1	50925.09	-1.82	648.75	7.99	8.03	59.42	43.58
Israel	522.83	31139.86	-0.67	474.88	1.8	26.83	70.78	70.44
Italy	20518.66	47753.26	-2.04	778.81	8.8	22.62	65.11	21.35
Jamaica	7378.89	35002.35	4.11	827.63	5.34	29.61	71.2	31.11
Japan	2647.24	15199.33	-0.29	233	2.8	9.27	56.07	89.81
Jordan	2087	50208.97	1.51	1424.45	1.68	12.54	83.93	87.03
Kazakhstan	21244.06	25094.45	-1.12	969.54	13.94	17.15	70.01	34.92
Kenya	20007.05	18419.52	0.56	616.89	7.61	8.03	54.02	64.35
Kiribati	7505.11	55552.64	-1.3	697.66	7.99	2.5	61.78	21.96
Kuwait	1648.69	94054.25	0.55	1174.32	3.82	20.76	76.61	63.01
Kyrgyzstan	12261.51	39595.41	0.84	1347.82	12.46	28.99	54.35	75.05
Laos	24396.91	17872.15	-2.27	1074.63	8.62	25.41	54.89	80.78
Latvia	5243.11	16216.39	-1.19	1181.14	9.98	16.17	62.56	88.07
Lebanon	10204.18	77723.67	1.83	546.58	6.94	9.74	76.17	54.74
Lesotho	6038.3	85539.92	1.22	788.16	13.6	19.1	54.09	93.65
Liberia	16320.78	32348.11	-1.47	1151.34	9.3	16.47	81.29	80.19

Libya	3944.4	30164.22	-0.27	1078.72	0.5	17.53	76.69	88.03
Liechtenstein	8894.78	78162.44	-1.78	1227.36	1.91	12.52	77.91	23.34
Lithuania	5961.31	68836.19	4.92	929.67	10.41	16.74	58.81	40.77
Luxembourg	4722.36	86376.04	3.42	581.24	6.93	28.47	55.37	62.17
Madagascar	13153.6	58398.99	-2.8	1264.58	13.98	17.39	74.38	92.1
Malawi	18388.5	15169.18	3.34	879.74	6.36	22.36	82.7	92.38
Malaysia	11722.37	11467.02	7.83	1216.4	1.87	27.7	80.45	56.18
Maldives	15373.57	38386.05	-2.4	486.04	12.04	1.13	61.67	45.44
Mali	13972.75	87450.4	0.81	503.09	11.06	14.11	57.86	50.27
Malta	3663.14	17415.65	2.48	607.45	13.72	11.51	70.32	66.27
Marshall Islands	341.44	63305.22	-1.04	1393.55	2.23	13.02	52.99	98.72
Mauritania	13057.57	56885.27	-2.26	1087.45	3.15	27.04	57.18	26.97
Mauritius	951.25	45268.71	3.21	95.29	11.63	14.15	68.35	49.23
Mexico	10420.44	53518.12	-1.29	263.8	12.93	28.44	63.07	34.1
Micronesia	16744.34	39302.76	-2.72	226.43	10.74	20.11	50.95	29.76
Moldova	6008.71	64092.3	-2.78	150.97	12	6.18	72.85	31.2
Monaco	2586.38	23706.82	4.94	1240.76	12.45	12.52	73.38	28.24
Mongolia	7622.55	85234.83	-2.86	124	3.12	1.77	56.35	61.89
Montenegro	10957.62	84889.68	5.99	495.64	3.89	12.01	70.66	33.86
Morocco	16228.25	39366.58	3.07	632.39	4.42	28.51	76.73	22.47
Mozambique	22580.3	46716.02	6.84	1159.79	6.38	1.65	59.4	58.21
Myanmar	16470.8	25093.02	-1.47	1210.65	14.77	16.25	56.01	34.24
Namibia	479.14	86926.95	-1.7	835.95	4.11	17.07	72.8	83.85
Nauru	5367.75	1835.81	-1.49	1305.03	13.11	18.32	71.02	69.19
Nepal	4560.48	86937.61	1.61	555.56	7.78	2.36	55.82	75.68
Netherlands	2153.68	57616.93	-0.3	564.48	4.33	11.31	75.17	36.44
New Zealand	14726.95	45643.95	4.3	1358.4	10.99	7.23	51.09	33.34
Nicaragua	15472.43	5644.31	2.46	865.43	5.01	23.36	53.73	16.69
Niger	18933.18	47475.28	4.57	630.51	3.7	24.75	77.98	71.83
Nigeria	7076.5	56399.73	0.97	132.8	13.76	4.97	83.26	49.69
North Korea	4814.27	51847.07	6.6	1061.73	12.1	20.1	74.23	85.58
North Macedonia	6492.12	46903.83	-0.57	1432.12	14.16	2.14	74.7	92.35
Norway	4695.78	54300.44	7.07	49.23	10.46	9.62	82.35	96.42
Oman	24550.98	45470.98	6.48	1224.6	4.79	25.04	51.3	63.07
Pakistan	5981	12157.4	-2.15	1009.62	5.1	22.02	52.29	38.06
Palau	14027.23	75286.12	0.51	907.55	13.29	18.86	58.15	12.17
Panama	22622.7	2803.58	6.62	766.96	14.09	24.16	84.93	41.21
Papua New Guinea	19947.13	38661.89	2.28	909.89	13.11	29.54	76.89	47.18
Paraguay	10955.86	70280.25	-0.37	160.2	5.32	9.33	60.37	30.79
Peru	1095.38	2483.72	7.86	620.28	5.76	20.71	57.64	94.55
Philippines	20445.18	9222.52	1.59	1274.72	14.17	14.55	71.47	24.87
Poland	25770.39	22623.47	7.37	941.99	9.12	15.87	58.07	25.71
Portugal	5733.42	18362.48	5.58	507.69	0.87	29.1	80.93	92.57
Qatar	25867.61	17180.93	1.36	1099.45	10.44	5.46	78.55	29.98
Romania	5820.03	51382.99	3.52	841.13	1.37	26.45	59.3	21.53
Russia	23107.56	90822.37	6.48	1173.8	9.83	16.97	53.04	46.35
Rwanda	9690.53	25268.81	4.96	719.02	1.22	7.39	73.91	16.78
Saint Kitts and Nevis	22131.53	47442.8	2.29	859	12.37	11.09	73.73	60.35
Saint Lucia	6943.47	83566.94	5.77	954.76	12.76	26.15	74.79	84.49

Saint Vincent and the Grenadi	18134.56	64869.26	3.8	1091.44	2.38	26.55	80.51	12.6
Samoa	21471.41	12939.54	0.69	1078.09	2.41	24.72	79.12	55.16
San Marino	167.03	27838.99	3.79	1422.72	9.48	8.53	72.19	58.06
Sao Tome and Principe	20276.2	10877.58	5.37	784.84	14.44	10.91	72.14	92.95
Saudi Arabia	2666.15	89086.92	4.57	98.37	4.51	21.54	52.36	61.81
Senegal	8993.61	59290.24	-2.5	1263.73	14.6	29.1	76.24	21.58
Serbia	19715.08	3116.09	-2.76	469.24	7.33	23.34	73.92	49.69
Seychelles	7115.02	94729.13	1.69	654.52	2.45	24.05	74.28	29.65
Sierra Leone	2142.82	64903.03	4.2	396.23	14.26	5.38	65.13	93.98
Singapore	10913.49	60949.14	1.37	397.62	14.76	12.87	81.29	30.47
Slovakia	5541.51	3732.83	4.17	534.37	12.97	14.72	83.89	26.51
Slovenia	22584.33	73955.42	5.48	1224.94	11.42	19.16	54.59	12.89
Solomon Islands	23942.12	58888.46	5.76	698.21	1.76	4.63	73.99	48.3
Somalia	5214.44	47108.21	-2.29	843.86	4.03	24.13	60.86	50.51
South Africa	303.12	7624.5	1.32	695.92	9	9.46	74.32	86.55
South Korea	20276.35	4532.09	2.29	152.16	3.63	29.61	54.99	54.4
South Sudan	16072.43	66972.2	3.16	14.18	4.9	16.01	53.08	41.21
Spain	864.25	8202.09	1.37	192.45	8.51	20.99	78.02	27.81
Sri Lanka	4355.38	10650.29	4	1024.39	0.47	28.15	51.82	58.18
Sudan	18435.86	82845.29	4.85	1162.51	5.09	24.63	52.8	89.64
Suriname	14237.85	77789.45	1.98	933.19	7.9	22.22	52.86	15.37
Sweden	6425.44	15829.11	6.59	317.87	14.64	10.77	56.37	80.28
Switzerland	17126.74	47730.04	3.11	1042.85	3.43	29.89	84.12	67.88
Syria	5188.9	64877.5	-2.21	44.46	3.87	14.42	80.39	74.72
Tajikistan	19310.63	40881.47	0.81	538.01	14.81	2.16	80.35	61.5
Tanzania	11404.56	69119.27	2.35	1266.46	13.51	13.23	59.69	62.72
Thailand	23721.53	20644.38	3.85	915.77	11	4.82	75.05	90.9
Timor-Leste	4672.58	23176.58	7.69	262.42	12.82	15.28	58.65	87.5
Togo	11578.49	49295.79	0.95	859.78	2.45	12.34	83.93	32.97
Tonga	17075.5	31432.9	5.51	189.78	14.55	14.16	58.26	16.54
Trinidad and Tobago	4414.54	49762.71	0.71	1201.88	6.46	8.21	71.6	72.9
Tunisia	4343.92	16589.73	-2.6	1067.79	9.96	14.76	79.55	81.7
Turkey	15219.63	82591.15	-0.74	162.29	4.05	2.66	68.59	93.36
Turkmenistan	1023.89	12302.75	1.97	1354.12	4.74	15.71	51.46	23.2
Tuvalu	25652.4	91714.18	-2.95	1380.13	9.59	26.17	65.92	55.89
Uganda	12710.52	63618.61	-1.46	43.47	4.62	21.44	57.06	69.94
Ukraine	25217.74	9645.45	4.4	643.44	13.02	6.14	74.24	84.59
United Arab Emirates	24560.03	65161.96	2.47	895.88	13.03	17.55	51.06	92.85
United Kingdom	17928.01	64527.56	-0.63	955.39	5.91	19.89	53.73	68.55
United States	25984.76	5341.57	7.75	590.02	13.06	23.69	69.85	75.72
Uruguay	22841.53	38870.02	0.6	968.01	12.12	23.11	77.92	48.77
Uzbekistan	21263.87	12123.69	2.99	8.36	4.87	11.63	63.87	71.9
Vanuatu	10103.12	43066.94	-0.39	541.22	3.41	3.12	71.12	69.47
Vatican City	16107.13	44461.14	1.18	1251.84	7.79	14.9	50.9	40.37
Venezuela	9885.71	38369.11	3.38	773.73	9.12	23.18	78.45	73.91
Vietnam	24843.66	2517.51	-0.85	10.98	9.71	27.04	58.52	92.51
Yemen	1567.89	88823.87	0.87	147.07	7.29	8.45	59.97	37.35
Zambia	20878.87	51588.99	0.42	884.99	10.74	8.91	64.47	20.85
Zimbabwe	4710.7	64961.3	-1	761.49	10.64	4.1	69.86	32.83

Deep Learning Model

For this project the Multi-Layer Perceptron (MLP) deep learning model was used in developing Python code that calculates and classifies countries based on their income. The MLP model is always layered with atleast three layers of perceptrons that form an interconnected neural network that accept data through multiple pathways to learn patterns. In this project, there are 3 layers of Perceptrons as shown in the image below:

```
# Build Deep Learning Model
model = Sequential()
model.add(Dense(16, activation='relu', input_dim=8))
model.add(Dense(8, activation='relu'))
model.add(Dense(1, activation='sigmoid'))
```

The LLM generated a deep learning model with 16, 8, 1 perceptrons per each of the three layers. The layers are categorized as Dense to signify that all input neutrons are connected to all neutrons in the next layer. In this case, we can see that the layers with 16 and 8 perceptrons are hidden layers since they have the same activation key and they require more perceptrons in their to weigh different economic and social indicators when making predictions.

Results

```
Epoch 45/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 1.0000 - loss: 0.0247
Epoch 46/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 1.0000 - loss: 0.0232
Epoch 47/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 1.0000 - loss: 0.0218
Epoch 48/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 1.0000 - loss: 0.0205
Epoch 49/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 1.0000 - loss: 0.0193
Epoch 50/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 1.0000 - loss: 0.0182
high_income
1    97
0    96
Name: count, dtype: int64
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step
Accuracy: 0.9230769230769231
Confusion Matrix:
[[17  1]
 [ 2 19]]
```

After running the CSV dataset that was generated by ChatGPT through the Python deep learning model, it outputed the results in the screenshot above. The Python code's deep learning model ran fifty times as shown by the last five of the fifty epochs, with each epoch being an indication of every time the MLP ran the dataset through its training model. After each epoch run, the accuracy for each run gradually increases until it reaches multiple runs with 100% training accuracy. Following the epochs, the model outputted an accuracy of approximately 0.923 or 92.3%, showing that the model correctly classified almost all of the countries in the dataset. The MLP model also outputted a confusion matrix, displaying the amount of true negatives (17), false positives (1), false negatives (2), and true positives (19) that were generated by the deep learning model in this run. Above the overall accuracy and confusion matrix, I asked the LLM to generate code that displays the total countries that fall under each binary classification of 1 or 0.

Challenges

There were a few challenges I faced through the completion of this project. I first had difficulties with coming up with instructions for ChatGPT to use in producing an output for my binary classification prompt from my earlier homework. I tested different instructions and decided upon prompting ChatGPT to use deep learning to create a Python model that is able to determine if a country can be classified as high-income or not, based on the classification problem prompt shown above. Another difficulty I faced in this project was figuring out how to create a Github repository that would be able to store the different files needed for this project. After troubleshooting and looking up online tutorials, I figured out how to use the website and was able to successfully upload different files to the repository.

Lessons and Experience

There were many lessons I learned and valuable experience I gained upon completing this project. I was able to see how deep learning models can be applied in analyzing real-world datasets in identifying data patterns and making predictions. I learned about the Multi-Layer Perceptron deep learning model that was used to generate multiple social and economic determinants for almost every UN-recognized country. I was able to see the different layers of perceptrons generated by the LLM that were used to understand patterns of data through multiple interconnected neurons, as well as witnessing the different epoch training of each time the dataset was analyzed. From this project, I also learned how to generate data labels based on calculated scores through LLM generated capabilities, combining both deep learning and weighing metrics to determine outcomes. Overall, this project gave many key learning experiences which I can take with me to understand the fundamentals behind deep learning models, like MLP, in further proccessing data compared to regular machine learning capabilities.